# Muhammad Muizz Zafar

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**Database Developer | SQL Server | T-SQL | Stored Procedures | Query Optimization**

Database developer focused on SQL Server 2022 migration, RBAC and access control, compliance, and T-SQL performance tuning.

## Skills

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| Databases: | Microsoft SQL Server (2019/2022), T-SQL, Stored Procedures, Views, Triggers, Indexing |
| Database Engineering: | Database Migration, Query Optimization, Execution Plan Analysis, Schema Design |
| Tools/Languages: | SQL Server Management Studio (SSMS), Git, Linux, SQL, Python, Java |

## Education

**University of Ottawa** — Ottawa, ON
Bachelor of Applied Science, Software Engineering *GPA: 3.7/4.0* — Sept 2020 - Dec 2024

## Work experience

**Database Developer** — May 2024 - Present
*Elections Canada* — Ottawa, ON

- Executed enterprise database migration by upgrading **SQL Server 2019 to SQL Server 2022** across **DEVTEST, UAT, and PROD** environments, migrating **50M+ records**, database schemas, stored procedures, triggers, and security configurations with zero downtime.
- Refactored legacy SQL code to modern **T-SQL standards**, updating deprecated syntax and optimizing database objects including stored procedures, views, and functions.
- Improved database performance through **query tuning, index optimization, and execution plan analysis**, significantly reducing query execution times on high-volume datasets.
- Audited and cleaned legacy database access rights and aligned role-based access models to the upgraded SQL Server 2022 environment, in line with government compliance requirements.
- Managed **role-based database access control** by configuring user permissions, roles, and security policies across multiple databases in accordance with government compliance requirements.
- Implemented **database backup, restore, and recovery procedures** across environments to support system reliability and disaster recovery readiness.
- Developed a **Python automation testing framework** to validate migration success by comparing schema structures, row counts, and data integrity across legacy and upgraded systems.
- Built internal **data analytics dashboards** using **Python, Streamlit, and Plotly Dash** to surface database health, operational metrics, validation trends, and data quality issues for directorate and senior management.
- Assisted with **database monitoring and maintenance**, including performance diagnostics, log analysis, and troubleshooting database issues across enterprise systems.
- Coordinated version-controlled database deployments and release processes, improving traceability and reliability of production data infrastructure across shared environments.
- Collaborated in an Agile cross-functional team with cloud engineers to improve **distributed data workflows**, release quality, and production reliability.

**Full Stack Developer** — September 2023 - April 2024
*University of Ottawa* — Ottawa, ON

- Built and maintained **JAVA** REST APIs and optimized database queries to improve response times for a platform serving **20,000+ users**.
- Implemented automated **unit and end-to-end tests** using Cypress, Jasmine, and JUnit to ensure application reliability.
- Investigated and resolved production issues by debugging backend APIs and database queries, improving system reliability.

**Cloud Developer Intern** — May 2023 - August 2023
*Trend Micro Inc.* — Ottawa, ON

- Developed serverless monitoring workflows using **AWS Lambda, S3, and CloudWatch** to analyze **50,000+ artifacts per day** and support automated vulnerability scanning pipelines for high-volume cloud artifacts.
- Defined and maintained cloud infrastructure using **Golang**, **AWS CDK**, through **CloudFormation** to provision **50+ AWS resources** and improve repeatability of service deployments.
- Implemented concurrent processing logic in **GoLang** using goroutines to improve throughput and strengthen fault handling for **distributed cloud data processing** workloads.
- Integrated vulnerability scanning tools (**Anchore, Grype**) with AWS services to automate security checks and operational monitoring across **50,000+ cloud artifacts per day**.

**Cloud Developer Intern** — September 2022 - December 2022
*Trend Micro Inc.* — Ottawa, ON

- Built a serverless automation pipeline using **AWS Lambda, S3, and Athena** to process incoming cloud data events, query metadata, and trigger downstream updates, reducing manual operational effort.
- Configured **Splunk** and **AWS CloudWatch** monitoring, alarms, dashboards, and alerts to track **15+ services** and improve observability of production cloud infrastructure.
- Developed **Docker-based integration tests** and unit tests to improve reliability of distributed serverless data pipelines.